

3. 已知 $AC[1:10, 10:20]$ 的维数 $w=1$.

$A[i+1, j+1] := a \times b - 10 \Rightarrow$ 求 E 式:

$(*, a, b, t_1)$
 $(-, t_1, 10, t_2)$
 $(+, i, 1, t_3)$
 $(+, j, 1, t_4)$
 $(*, t_3, 11, t_5)$
 $(+, t_4, t_5, t_6)$
 $(-, A, 21, t_7)$
 $([] =, t_2, -, t_7, [t_6])$

自左而右的归约
 offset $C = (l_1 \times n_2 + l_2) \times w$
 CONS PART = $A-21$
 VAR PART = $t_1 \times n_2 + t_2$
 $= t_1 \times 11 + t_2$

交换位置
 $(1 \times 11 + 10) \times 1 = 21$

译本:

1. ① 第0层: a, b, p

1 : s, r

2 : v

② 第1层: a, b, p, q

2 : r, y

3. $S \rightarrow *E_1 := E_2$ { emit ($*E_1.place := E_2.place$); }

$S \rightarrow id := *E$ { $p := \text{lookup}(id.name);$
 if $p \neq nil$
 then emit ($id.place := *E.place$)
 else error
 }

$E \rightarrow E_1$ { ~~$E.place := \text{newtemp};$~~
 ~~$E_1.place;$~~

$E \rightarrow E_1$ { $E.place := \text{newtemp};$
 emit ($E.place := *E_1.place$);
 }

4. $E \rightarrow E_1 \uparrow E_2$ { ~~$E_{\text{place}} := \text{newtemp}$; $t_1.\text{place} := \text{newtemp}$;~~
 ~~$\text{emit}(t_1 := 'E_1' \text{ AND } E_2.\text{place})$;~~
 ~~$\text{emit}(E.\text{place} := 'NOT' t_1)$;~~
 }

{ ~~$\text{backpatch}(E_1.\text{true}, E_2.\text{codebegin})$;~~ 100 if E_1 goto 102
 ~~$E.\text{codebegin} := E_1.\text{codebegin}$;~~ 101 false goto 106
 ~~$E.\text{true} := \text{merge}($~~ 102 ~ E_2 goto 104
 ~~$E_1.\text{false}, E_2.\text{false})$;~~ 103 false goto 106
 ~~$E.\text{false} := E_2.\text{true}$;~~ 104 true
 } 105 goto 107
 106 false
 107

5. $S \rightarrow \text{repeat } S_1 \text{ until } E$

~~$\text{backpatch}($~~
 ~~$S.\text{codebegin} := S_1.\text{codebegin}$;~~
 ~~$\text{backpatch}(S_1.\text{chain}, E.\text{codebegin})$;~~
 ~~$\text{backpatch}(E.\text{false}, S_1.\text{codebegin})$;~~
 ~~$S.\text{chain} := E.\text{true}$;~~
 ~~$S.\text{codebegin} := S_1.\text{codebegin}$;~~
~~}~~

8. $S \rightarrow \text{for } (S_1 : E : S_2) S_3$

{ $S.\text{codebegin} := S_1.\text{codebegin}$;
 $S.\text{chain} := E.\text{false}$;
 $\text{backpatch}(S_3.\text{chain}, S_2.\text{codebegin})$
 $\text{backpatch}(S_2.\text{chain}, E.\text{codebegin})$
 $\text{backpatch}(S_1.\text{chain}, E.\text{codebegin})$
 $\text{backpatch}(E.\text{true}, S_3.\text{codebegin})$
 }

9. { backpatch (E_1 , false, E_2 .codebegin); 100 E_1 goto F
 backpatch (E_2 .true, E_3 .codebegin); 101 goto 102
 E .codebegin := E_1 .codebegin; 102 E_2 goto 104
 E .true := E_3 .false; merge(103 goto F
 E .false := merge (E_1 .true, (E_2 .false, 104 E_3 goto F
 E_3 .true)); } 105 goto T

10. 分析真值表 对于 $(E_1 \wedge \hat{E}_3) \vee (\hat{E}_1 \wedge E_2)$ 100 E_1 goto 104
 { backpatch (E_1 .true, E_3 .codebegin); 101 goto 102
 b (E_1 .f, E_2 .t); 102 E_2 T
 E .t := E_1 .t; 103 F
 E .f := m (E_2 .f, E_3 .t); 104 E_3 F
 E .t := merge (E_2 .t, E_3 .f); } 105 T

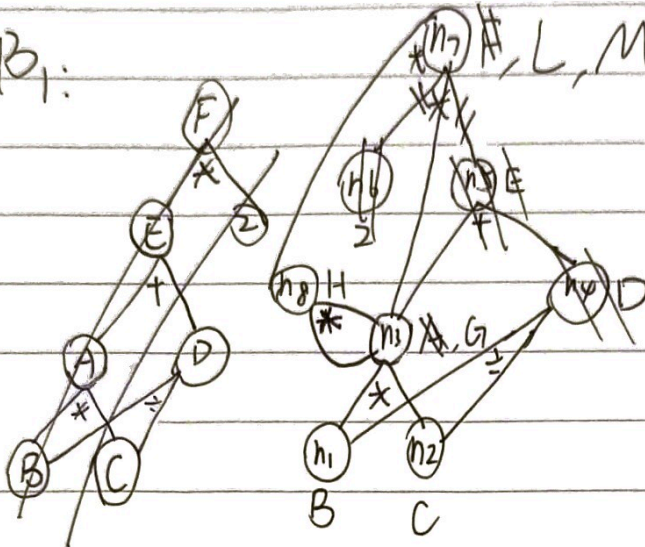
Chapter. 10.

1. 到中间代码. 时 & 空. 并做变换.

技术
局部优化:
编译
全局

4.

B₁:



$$G := B * C$$

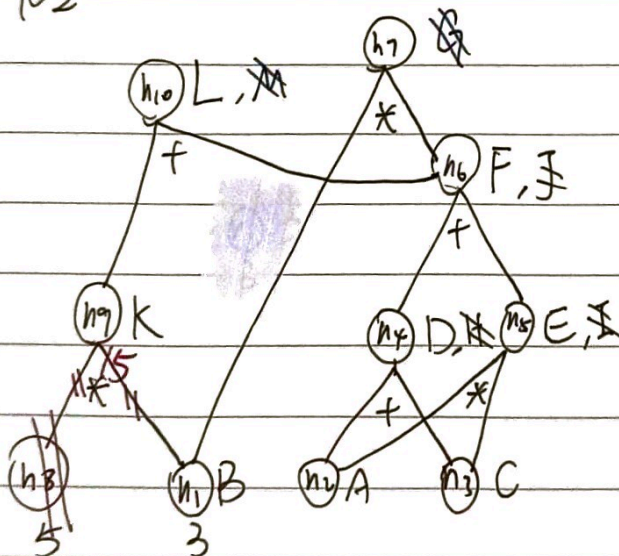
$$H := G * G$$

$$L := H * G$$

$$M := L$$

(2) 进说明去 "M:=L".

B₂:



$$(1). B := 3$$

$$D := A + C$$

$$D := A + C$$

$$E := A * C$$

$$E := A * C$$

$$F := D + E \Rightarrow$$

$$F := D + E$$

$$G := B * F$$

$$G := 3 * F$$

$$K := B * 5$$

$$L := K + F$$

$$L := 15 + F$$

$$M := L$$

$$M := L$$

7. 输入

指令选择
指令分配
指令调度

附:

No.

DATE / /

1. $B[0:5][2:4][1:4][4:8]$, each element 8 Byte, BL 编号 1000 的
字节地址求 $B[4][2][3][7]$ 的字节位置.

解:

$$VARPART = [(4 \times 3 + 2) \times 4 + 3] \times 5 + 7 = 302 \times 8 = 2416$$

$$C = [(0 \times 3 + 2) \times 4 + 1] \times 5 + 4 = 49 \times 8 = 392$$

$$CONSPART = B - C = 1000 - 392 = 608$$

$$\Rightarrow 608 + 2416 = 3024$$

$$B - C + VARPART$$

$$= B + (VARPART - C)$$

求地址

字节地址

$$= B + [((4 - 0) \times 3 + (2 - 2)) \times 4 + (3 - 1)] \times 8$$

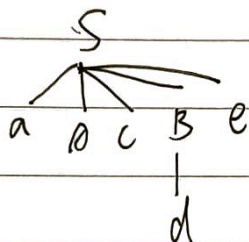
$$= B + 253 \times 8$$

$$= 3024$$

2. $S \rightarrow aAcBe$

$A \rightarrow Ab|b$

$B \rightarrow d$



求 $aAbcd e$ 的字节地址

$\{\epsilon, a, aA, aAc, aAcd\}$

$\cup \{aAcB, aAcBe\}$ 的字节地址

字节地址